# Naresh Modina

Machine Learning Engineer | AI/ML Researcher
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## SKILLS

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| **Programming** | Python (advanced), HTML/CSS (Intermediate). |
| **ML / DL** | PyTorch (advanced), TensorFlow(Intermediate), Scikit-learn(advanced), NumPy / Pandas (advanced). |
| **LLMs / NLP** | Hugging Face Transformers, Datasets, SFT, PEFT / LoRA. |
| **Reinforcement Learning** | DQN, PPO, Actor-critic, Gymnasium, World Models. |
| **MLOps** | Docker, Git, torchrun (multi-GPU DDP), distributed training, Conda, SQL. |
| **Soft Skills** | Technical writing, presentation, team coordination, research grant proposals. |

## CURRENT POSITION

**Applied AI Researcher** *2023 – 2026*
Conservatoire National des Arts et Métiers (Cnam), Paris, France.

## RESEARCH EXPERIENCE

**Projects**
**Tele-SLMs — Open Source Project** *2026*

- Explores whether sub-2B parameter SLMs can be specialized for 3GPP telecommunications standards — accurately and with low retraining overhead. Two-stage pipeline: continual pretraining on ~1.26B tokens (3GPP standards + arXiv), followed by instruction fine-tuning.
- Benchmarks SmolLM2 and Qwen2.5 sub-2B variants on Tele-Eval. Multi-GPU training via PyTorch DDP on 3× L40S.
- Curated a training-ready telecom NLP dataset, available on HuggingFace.

**NEXASPHERE — European Project** *2025 – 2027*
Development of monitoring and state assessment solutions for 3D networks integrating terrestrial and non-terrestrial (NTN) components towards 6G. Design of DRL-based solutions for automatic anomaly resolution via dynamic network reconfiguration. Partners include SAFRAN, HPE, OHB, HellaSat, Stellantis.

**INFLUENCE — BPIFrance Project** *2022 – 2026*
Design of the ARM (Autonomous Reconfiguration–Intent Management) framework to resolve performance anomalies in 5G/xG systems through automated, intent-based recovery. Used

DRL for intelligent container sizing and VNF resource management. Filed a patent in collaboration with Orange research team. Partners include Orange and Nokia.

**MAESTRO5G — ANR Project** *2019 – 2022*
Optimization of IoT data orchestration through Age of Information (AoI)-based pricing and Markov Decision Processes (MDP), reducing operational costs and designing 5G slicing frameworks with multi-level equity. Resulted in publications in IEEE TMC, ICC, and ITC. Partners include Orange, Nokia, Inria, CentraleSupélec, IPP.

### Supervision & Dataset Leadership

**5G3E Testbed** *2023 – 2026*
Led the development of an end-to-end 5G emulation testbed and directed the creation, curation, and publication of the 5G3E open-source dataset for AI/ML tasks in 5G networks.

### Funding Proposals

- NEXASPHERE (2024) — Contribution focused on AI-based resource management.
- iLab (2026) — Drafted a competitive research and innovation proposal outlining objectives, technical methodology, impact, and implementation roadmap.
- Pionnier de l'IA (2026) — Developed a comprehensive AI-focused funding proposal detailing innovative contributions, research strategy, and expected scientific and societal impact.

## EDUCATION

**PhD in Computer Science** *2019 – 2022*
Avignon Université, France — Thesis: Resource allocation in 5G wireless networks

**Master of Telecommunication Engineering** *2015 – 2019*
Politecnico di Milano, Italy

## TEACHING

**AI4CI — European Master Programme** *2024 – 2026*
Teaching Artificial Intelligence and Machine Learning for Connected Systems, demonstrating strong foundations across ML, DL, and RL applied to network systems.

## PUBLICATIONS & PATENT

- Patent: Procédés de détection et de gestion d'anomalies impactant un environnement informatique (FR3164046, 2024) — filed in collaboration with Orange.
- 4 peer-reviewed publications in IEEE TMC, ITC, and ICC on next-generation wireless networks. Full list available on Google Scholar.

## LANGUAGES

**English** (Professional fluency) | **French** (A2) | **Hindi** (Professional fluency) | **Telugu** (Native)